The Isolated-Dummy FIFO Linking Conjecture: Affine Response and the Causal Frontier

a9lim

Abstract

Let a finite simple graph be played by opening its vertices into a FIFO queue and later closing the queue front. Closing a vertex toggles a binary score exactly when it has odd degree into the untouched set. The isolated-dummy FIFO linking conjecture asserts that adjoining one isolated vertex lets either designated player force terminal score zero. The conjecture remains open.

We give a graph-free formulation of the problem. Every complete history has a disjointness vector in the binary edge space on the real vertices. For a fixed odd-seeking strategy, the zero-seeking responses are successful for every graph if and only if zero belongs to the affine hull of their terminal disjointness vectors. Equivalently, the responses must support an odd formal flow of zero edge moment. FIFO front deletion yields a canonical cut-space filtration, but prefix cuts and descendant continuation spaces remain correlated.

The paper records the proved part of this reduction: live-star and complete reply-fan identities, parity-cell and matching subclasses, complete exclusion of close-first counterstrategies, fixed-front contraction, and a strategy-relative minimal-bad theorem reducing every hypothetical counterstrategy to charged-CLOSE-spike or protected-singleton ancestry. Explicit families show that childwise contraction and uniformly bounded affine certificates are false. The remaining assertion is a causal factor-extension theorem coupling earlier defender siblings to descendant continuation cosets. We prove that three tempting globalizations cannot supply that theorem: the pruned history tree has no nonzero one-cycles, commuting OPEN/CLOSE squares carry a genuine unit-edge curvature, and erasing the dummy is not a map between fixed strategy trees because the required ko-wall exchange reverses the controller. A nine-real-vertex root fan in which eight of the nine replies lose shows how sharply the surviving same-parity selector is being used. An exhaustive census through eight real vertices plus the isolated dummy is consistent with the conjecture for both designated seats, but is not a universal proof.

1 Introduction

Queue layouts place graph edges under a nonnesting FIFO discipline [2]. The game studied here instead puts the *vertices* in a FIFO queue and lets play construct their activity intervals. Those intervals are proper—none properly contains another—in the standard interval-graph sense [3], but their endpoints are chosen adversarially. This distinction is decisive: the payoff depends on which graph edges join disjoint activity intervals, while the legal schedule does not depend on the graph.

Let R be a finite set of real vertices, let G be a simple graph on R, and adjoin an isolated vertex d. Two players alternate legal moves. One designated player, called the *defender*, seeks terminal score 0 in $\mathbb{F}_2$; the other, called the *attacker*, seeks score 1. Either physical seat may be designated defender. The central statement is the following.

Conjecture 1.1 (FIFO linking)**.** *For every finite simple graph G and isolated dummy d, the defender has a strategy forcing terminal score zero, whichever seat is designated defender.*

The conjecture is a strict combinatorial strengthening of the application that motivated it. In the Gold–Arf normal-play construction, a public Witt frame reduces the support graph to a matching plus isolates. That subclass is proved independently and does not require Conjecture 1.1. Nothing below upgrades the matching theorem to arbitrary graphs.

The purpose of this paper is to state the current theorem frontier. Sections 2– 4 develop the exact affine formulation. Section 5 collects the strongest proved regimes used by the reduction. Section 6 gives the candidate minimal obstruction and the remaining causal obligations. Section 7 separates formal proof from finite computation.

2 The FIFO parity game

Put $V = R \sqcup \{d\}$ and extend G by declaring d isolated. A state is

$$(U, Q, k, t, g),$$

where $U \subseteq V$ is the untouched set, Q is an ordered list of distinct open vertices, $k \in \{0, 1\}$ is the ko bit, t is the player to move, and $g \in \mathbb{F}_2$ is the accumulated score. Reachable states satisfy $U \cap Q = \varnothing$.

Definition 2.1 (moves)**.** The legal transitions are as follows.

1. OPEN(x), for $x \in U$, removes x from U, appends it to Q, and sets $k = 1$ exactly when the old queue was empty.
2. CLOSE is legal when $Q = (f, Q_0)$ and $k = 0$. It removes f and changes the score by

$$\chi_G(f, U) = \deg_G(f, U) \pmod 2.$$

3. If neither an open nor a close is legal, a forced pass clears k and changes nothing else.

Every transition changes the player to move. The initial state is $(V, (), 0, t_0, 0)$, and a state is terminal when $U = \varnothing$ and $Q = ()$.

On a reachable state, ko is set only on a singleton queue created by opening onto an empty queue. Consequently a forced pass occurs only after $U = \varnothing$; no subsequent close can score. The rank

$$4|U| + 2|Q| + k$$

strictly decreases on every open, close, and forced pass. Thus the game tree is finite and ordinary backward induction gives determinacy for either designated target.

For a complete history h, write o_x and c_x for the times at which x opens and closes, and put $I_x = [o_x, c_x]$. FIFO makes the opening and closing orders identical. Hence the intervals I_x are pairwise nonnesting.

Let

$$\mathcal{E}_R = \mathbb{F}_2^{\binom{R}{2}}$$

be the binary vector space with one coordinate xy for each unordered real pair. Define the *disjointness vector*

$$D(h) = \sum_{\{x,y\} \subseteq R} [I_x \cap I_y = \varnothing] \, xy \in \mathcal{E}_R.$$

If $A_G \in \mathcal{E}_R$ is the adjacency vector of G, use the standard coordinate pairing $\langle -, - \rangle$.

Proposition 2.2 (interval accounting)**.** *For every complete history,*

$$g(h) = \langle A_G, D(h) \rangle.$$

Proof. Fix an edge xy and suppose x opens before y. The edge contributes to a close charge exactly when x closes while y is untouched. By FIFO this is equivalent to $c_x < o_y$, hence to $I_x \cap I_y = \varnothing$. Summing over the edges of G proves the identity. □

The graph affects only the final linear functional. Move legality and the vector $D(h)$ depend solely on the schedule.

3 Affine response

Fix an attacker seat and a deterministic attacker strategy S in the schedule tree. The strategy may depend on the complete history. Let $\mathcal{H}_S$ be the set of terminal histories compatible with S and with arbitrary defender choices.

Theorem 3.1 (affine separation)**.** *The strategy S forces score one for some graph on R if and only if*

$$0 \notin \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}.$$

Consequently, Conjecture 1.1 is equivalent, for both attacker seats, to

$$0 \in \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\} \qquad \textit{for every deterministic } S.$$

Proof. If S forces score one on G, Proposition 2.2 places every $D(h)$ in the affine hyperplane $\langle A_G, z \rangle = 1$, which excludes zero. Conversely, write the affine hull as $a + W$. If it excludes zero, then $a \notin W$, so linear separation over $\mathbb{F}_2$ gives a functional ℓ with $\ell(W) = 0$ and $\ell(a) = 1$. Every functional on $\mathcal{E}_R$ is pairing with the adjacency vector of a simple graph on R. For that graph, S forces score one. □

Over $\mathbb{F}_2$, membership in the affine hull has an especially concrete form.

Corollary 3.2 (odd response flow)**.** *The affine condition in Theorem 3.1 holds for S if and only if there is an odd-cardinality formal family of compatible terminal histories whose disjointness vectors sum to zero.*

At attacker nodes all incoming flow follows the move selected by S; at defender nodes it may split among an odd number of children. This is an algebraic response flow, not a randomized strategy.

If affine contraction fails, Theorem 3.1 supplies a separating graph. On the finite public-state graph for that fixed scalar game, backward induction lets the attacker choose one winning child at each attacker node. Thus a counterexample may be taken positional in the full state (U, Q, k, t, g). The score coordinate cannot be dropped: two schedule paths may reconverge publicly on opposite score sheets.

3.1 Universal move vectors

For a live real set L and $x \in L$, define its complete-graph star

$$s_L(x) = \sum_{y \in L \setminus \{x\}} xy \in \mathcal{E}_R.$$

The dummy star is zero. Give an open of x the vector $s_L(x)$, measured just before the open, and give closes and passes vector zero.

Proposition 3.3 (live-star identity)**.** *For every complete history,*

$$D(h) = \sum_{x \text{ opened}} s_{L_x}(x),$$

where L_x is the real live set when x opens.

Proof. Suppose x opens before y. The coordinate xy appears once when x opens. It appears a second time when y opens exactly if x is still live, that is, exactly if the intervals overlap. The two copies then cancel. If the intervals are disjoint there is no second copy, so the displayed sum has the coordinate prescribed by $D(h)$. ☐

There is a dual close-based description. A close of front f while the real untouched set is U contributes the cut star

$$c(f, U) = \sum_{u \in U} fu.$$

The sum of these close vectors is also $D(h)$. Evaluating either description on A_G gives Proposition 2.2.

3.2 The complete reply fan

For a state with untouched set U and queue Q, define the scalar queue cut

$$P_G(Q, U) = e_G(Q, U) \pmod 2.$$

Closing the front changes both P_G and the accumulated score by the same bit. Opening a vertex x changes P_G by the degree of x in the current live graph.

Fix $x \in U$. Let P_x be the queue cut after $\text{OPEN}(x)$, and let P_{xy} be the cut after the two successive opens x, y.

Proposition 3.4 (complete pair-response identity)**.** *For every well-formed state,*

$$\sum_{y \in U \setminus \{x\}} P_{xy} = |U| P_x \quad \text{in } \mathbb{F}_2.$$

If $|U|$ is even, the $|U| - 1$ second-open replies are an odd family with zero aggregate cut. If $|U|$ is odd and the old queue is nonempty, adjoining the now-legal front close, translated back by its score charge, again gives an odd family with zero aggregate cut.

Proof. Opening does not change the live set. If $\deg_L$ denotes live degree, then

$$P_{xy} = P_G(Q,U) + \deg_L(x) + \deg_L(y).$$

Handshaking on the live graph gives

$$\sum_{u \in U} \deg_L(u) = P_G(Q,U).$$

Summing the first identity over $y \neq x$ and eliminating the full degree sum gives the displayed identity. For odd $|U|$, the open family has even cardinality and sum P_x; the translated close representative contributes the same P_x. $\square$

The sole nonterminal local exception is an odd untouched set with an empty queue: the first open sets ko and there is no close representative. At the isolated-dummy root this is precisely the protected universal fan. More importantly, Proposition 3.4 concerns prefix cuts. Below each reply the fixed attacker strategy supplies a generally different continuation coset. Transporting the prefix cancellation through those cosets is the unresolved issue.

3.3 Continuation cosets

At a node q of one fixed attacker strategy, let Λ_q be its compatible terminal leaves. For $\epsilon \in \mathbb{F}_2$, define

$$\mathcal{C}_q^\epsilon = \left\{ \sum_{\lambda \in \Lambda_q} e_\lambda D_q(\lambda) : \sum_\lambda e_\lambda = \epsilon \right\},$$

where $D_q(\lambda)$ is the future edge moment from q to λ. Then $W_q = \mathcal{C}_q^0$ is a linear space and $A_q = \mathcal{C}_q^1$ is a nonempty affine W_q-coset.

Proposition 3.5 (parity convolution)**.** *Suppose a defender node q has children q_m and move vectors c_m. Choose $a_m \in A_{q_m}$ and fix one child m_0. Then*

$$\begin{aligned} W_q &= \sum_m W_{q_m} + \operatorname{span}\{(c_m + a_m) + (c_{m_0} + a_{m_0}) : m \neq m_0\}, \\ A_q &= c_{m_0} + a_{m_0} + W_q. \end{aligned}$$

At an attacker node with selected child q' and move vector c, $W_q = W_{q'}$ and $A_q = c + A_{q'}$.

Proof. A formal leaf family splits uniquely among the disjoint child leaf sets. Its augmentation and moment are the sums of the child augmentations and translated child moments. Every odd child-parity profile is obtained from the profile supported on m_0 by toggling pairs of child parities. Internal even corrections give $\sum_m W_{q_m}$; toggling m_0 together with m gives the displayed generators. This proves both formulas. $\square$

Let $\mathcal{F}$ be an antichain meeting every terminal branch below a root exactly once. Write Γ_q for the prefix moment from the root to $q \in \mathcal{F}$. Iterating Proposition 3.5 gives

$$A_{\text{root}} = \operatorname{Aff}\{\Gamma_q + a_q : q \in \mathcal{F}\} + \sum_{q \in \mathcal{F}} W_q.$$

This frontier decomposition is the exact place where prefix cuts and continuation directions meet.

4 The FIFO cut filtration

Let $L \subseteq R$ be the current live real set and let $f \in L$ be the FIFO front. Write

$$\mathrm{Cut}(L) = \mathrm{span}\{s_L(x) : x \in L\} \subseteq \mathcal{E}_L.$$

Theorem 4.1 (front splitting)**.** *There is a canonical direct sum*

$$\mathcal{E}_L = \mathrm{Cut}(L) \oplus \mathcal{E}_{L \setminus \{f\}}.$$

The projection to the second summand is

$$(T_f z)_{ij} = z_{ij} + z_{fi} + z_{fj}, \qquad i, j \neq f.$$

It kills every live star and is the identity on vectors supported away from f.

Proof. The formula for T_f kills $s_L(f)$ because its two incident terms cancel. For $x \neq f$, the ij and fi coordinates of $s_L(x)$ cancel. It plainly fixes $\mathcal{E}_{L \setminus \{f\}}$. The only relation among the live stars is $\sum_{x \in L} s_L(x) = 0$: in any relation, the coordinate ij says that the coefficients of $s_L(i)$ and $s_L(j)$ agree. Hence $\dim \mathrm{Cut}(L) = |L| - 1$, and finally

$$\dim \mathrm{Cut}(L) + \dim \mathcal{E}_{L \setminus \{f\}} = (|L| - 1) + \binom{|L| - 1}{2} = \binom{|L|}{2}.$$

□

If z is regarded as a simplicial 1-cochain on the full simplex, then

$$(\delta z)_{fij} = z_{fi} + z_{fj} + z_{ij} = (T_f z)_{ij}.$$

Thus T_f is the chart at f of the associated two-graph; this is the standard switching/coboundary correspondence [4]. Successive front charts satisfy the absorption identity $T_g T_f = T_g$ after both fronts are deleted.

While f stays at the queue front, every open contributes a current live star and therefore lies in the first summand of the displayed direct sum. After f closes, every later contribution lies in the smaller edge space. FIFO therefore gives a triangular filtration by successive front deletions. The dummy is exceptional: its star is zero, but closing it does not reduce the real edge space.

The filtration does not by itself prove contraction. A prefix cut can cancel only together with a continuation representative from the same strategy branch. Projecting first and choosing continuations later silently replaces the correlated set in the frontier decomposition by a larger Cartesian product. Explicit examples in Section 6 show that this replacement is false.

The correlation has an exact local form. Fix a real front f. For a terminal continuation λ, let $p_f(\lambda)$ be the indicator of the real vertices still untouched when f closes, and let $W_f(\lambda)$ be the part of its relative disjointness vector supported away from f. If δp denotes the cut with coordinates $(\delta p)_{ij} = p_i + p_j$, then

$$D_f(\lambda) = \sum_i p_f(\lambda)_i \, fi + W_f(\lambda), \qquad T_f D_f(\lambda) = W_f(\lambda) + \delta p_f(\lambda).$$

By the direct sum in Theorem 4.1, every formal leaf family c satisfies

$$D_f(c) = 0 \quad \Longleftrightarrow \quad p_f(c) = 0 \ \text{ and } \ W_f(c) = 0.$$

Thus front projection replaces one edge moment by a joint cut/continuation moment; it does not decouple the two coordinates.

One half of this joint target contracts at the initial front.

Proposition 4.2 (initial cut contraction). *Suppose the attacker opens a real initial front f, leaving the isolated dummy untouched. For every fixed attacker strategy up to the close of f, zero lies in the affine hull of the compatible vectors $p_f(\lambda)$.*

Proof. If not, a linear functional with vertex weights a_i would take value one on every compatible remainder. Give the dummy weight zero, and let s be the total weight of the vertices still available to open. On the ko-protected first response, the defender opens the dummy when $s = 0$ and a weight-one real vertex when $s = 1$; either choice leaves total available weight zero. Thereafter, whenever the defender moves before f closes, it closes f if $s = 0$ and otherwise opens a weight-one vertex. If the attacker opens a weight-one vertex from total zero, a second such vertex remains because their number was positive and even. Hence $s = 0$ whenever the attacker closes f, contradicting the separating functional. □

5 Proved regimes and reductions

5.1 Parity cells and matchings

Theorem 5.1 (parity-cell pairing). *Suppose V has a perfect partition $\mathcal{M}$ into two-element cells such that*

$$e_G(A, B) = 0 \pmod 2 \qquad (A, B \in \mathcal{M},\ A \neq B).$$

Then the second player forces terminal score zero.

Proof. When the first player opens one member of a cell, the second opens its mate. When the first closes the front member of a cell, its mate is the next front and the second closes it. Hence the untouched set is always a union of cells and the queue is a concatenation of cells. Ko causes no exception, because the response to an open onto the empty queue is the mate open.

If $A = \{a, a'\}$ closes while the untouched set is a union of cells, its two charges sum to

$$\deg_U(a) + \deg_U(a') = \sum_{B \subseteq U} e_G(A, B) = 0.$$

Every close belongs to one such pair. □

This condition strictly extends literal twin pairing but is not universal. The separately proved matching-plus-isolates theorem is stronger in a different direction: it gives either-seat strategies, without requiring an isolated dummy, for every matching graph. That theorem is the one used by the Gold–Arf construction; Theorem 5.1 and the matching theorem do not settle arbitrary support graphs.

The first cell may also be balanced by a two-bit form of the handshake lemma. For a finite graph H, put

$$p_v = \deg_H(v) \pmod 2, \qquad b_v = \sum_{u \sim_H v} p_u \pmod 2, \qquad c(v) = (p_v, b_v).$$

Lemma 5.2 (two-bit handshake). *If H has even order, every fibre of $c\colon V(H) \to \mathbb{F}_2^2$ has even cardinality. Hence every vertex has a distinct vertex of the same colour. Moreover, if $c(v) = c(w)$ and $A = \{v, w\}$, then each fibre of*

$$x \longmapsto \big(p_x, e_H(x, A) \bmod 2\big), \qquad x \in V(H) \setminus A,$$

also has even cardinality.

Proof. Let $O = \{v : p_v = 1\}$. Handshaking gives $|O| = 0$ in $\mathbb{F}_2$, while $\sum_{v \in O} b_v = 2e_H(O) = 0$. Thus both b-fibres in O are even. Furthermore $e_H(O, V(H) \setminus O) = \sum_{v \in O} \deg_H(v) = |O| = 0$, so the same is true in the complement. For the second assertion, the four moments of the pair (p_x, r_x), where $r_x = e_H(x, A)$, vanish:

$$\sum_x 1 = \sum_x p_x = \sum_x r_x = \sum_x p_x r_x = 0.$$

The last two identities are respectively $p_v + p_w = 0$ and $b_v + b_w = 0$; the possible edge vw cancels because $p_v = p_w$. The four fibre indicators are the four polynomials $(1+p)(1+r), (1+p)r, p(1+r), pr$. $\square$

If an attacker opens v, opening a same-colour w matches the first pair's close charges. If the attacker instead opens another vertex x, the second part of Lemma 5.2 supplies a compatible second mate. This is a two-round refinement, not an induction: later play may split the balanced fibres and require a close response.

5.2 Fixed-front contraction

The following elementary game is the local prefix model at a fixed real front y. Let S be the vertices which may still be opened before y closes. A leaf is labelled by the remainder $T \subseteq S$ left untouched at that close.

Theorem 5.3 (fixed-front prefix contraction)**.** *From a defender-controlled root, for every deterministic attacker policy the affine hull of the attainable remainder indicators contains zero.*

Proof. It is enough to exclude affine separation. Let ℓ be a linear functional on $\mathbb{F}_2^S$. For $T \subseteq S$, let $A(T)$ mean that an attacker to move can force ℓ to equal one at the eventual close of y, and let $D(T)$ denote the same assertion with the defender to move. Before y closes, an open only deletes its label from T, so

$$\begin{aligned} A(T) &\iff \ell(T) = 1 \quad \text{or} \quad \big(\exists w \in T : \ D(T \setminus \{w\})\big), \\ D(T) &\iff \ell(T) = 1 \quad \text{and} \quad \big(\forall v \in T : \ A(T \setminus \{v\})\big). \end{aligned}$$

Simultaneous induction on $|T|$ gives $D(T) = \text{false}$ and $A(T) \iff \ell(T) = 1$. Indeed, the smaller D terms vanish. If $D(T)$ held, then $\ell(T) = \ell(T \setminus \{v\}) = 1$ for every $v \in T$. Writing $a_v = \ell(\{v\})$ would give $a_v = 0$ for every v and also $\ell(T) = 1$, a contradiction. Hence no affine hyperplane separates every defender response leaf from zero. $\square$

The distinguished immediate-close leaf cannot in general be prescribed in the affine relation. Thus Theorem 5.3 contracts the absolute prefix, not the relative spine needed after a particular charged close has been selected.

One exact endgame corridor permits the required change of action type. Write $B(u, v) \in \mathbb{F}_2$ for adjacency.

Lemma 5.4 (two-switch tail)**.** *Suppose the defender moves just after an attacker close, the residual score target is zero, and*

$$U = \{x, y\}, \qquad Q = (v_1, \ldots, v_{2r+1}).$$

If

$$B(v_{2i-1}, y) = B(v_{2i}, y) \quad (1 \le i \le r), \qquad B(v_{2r+1}, y) = B(x, y),$$

then the defender forces future score zero with at most two changes between opening and closing relative to the attacker's preceding move.

Proof. Open x. The even queue is then paired as $(v_1, v_2), \dots, (v_{2r+1}, x)$, and the two members of each pair have equal adjacency to the sole untouched vertex y. While the attacker closes, close the next front; each pair contributes equal charges. If the attacker opens y, close the front, after which every remaining close has charge zero. If y is never opened before the pairs drain, it is the harmless terminal singleton. Only the initial close/open and the possible later open/close are action-type switches. $\square$

5.3 Close-first attackers

Call an attacker *close-first* if it closes whenever the queue is nonempty and ko is clear, and opens only when a close is unavailable.

Theorem 5.5 (conditioned close-first contraction)**.** *Let the defender move from a well-formed state with nonempty queue and clear ko. Against a close-first attacker, the defender can force future score increment zero. No dummy and no parity hypothesis are required.*

Proof. Assume a counterexample of minimum remaining rank and write the queue as (f, Q_0). The cases $|U| \le 1$ drain directly. For every $x \in U$, let the defender open x; the attacker closes f. Minimality forces

$$\deg_{U \setminus \{x\}}(f) = 1 \qquad (x \in U),$$

since a zero charge would leave a smaller counterexample. Therefore f is adjacent to every member of U, $|U|$ is even, and $\deg_U(f) = 0$.

If $Q_0 = ()$, close f at charge zero. The attacker's next open creates a ko-protected singleton; a second open makes the first new front available. Applying the same minimum condition to all choices forces a vertex to be simultaneously adjacent and nonadjacent to that front, a contradiction. If $Q_0 = (h, Q_1)$, compare the prefixes

$$\mathrm{OPEN}(x), C_f, \mathrm{OPEN}(y), C_h \qquad (x \neq y).$$

Minimality forces every twice-punctured h-degree to vanish. Comparing pairs with one common vertex makes the h-row constant on U; even $|U|$ then gives $\deg_U(h) = 0$. Closing f, h reaches a smaller counterexample, possibly after the two ko-clearing opens when $Q_1 = ()$. This is impossible. The remaining two-point case is the same calculation without a third vertex: opening either point and then draining the original queue gives two equations whose sum makes the direct drain neutral. $\square$

At an empty root one needs a stopped version because a normalization argument may expose score one before termination. Permit a close-first attacker to stop and win at a clear score-one node. The defender still has all ordinary moves.

Theorem 5.6 (stopped empty-root contraction)**.** *From a score-zero empty-queue root with the defender to move, no stopped close-first attacker can force a stop or terminal score one, on any finite graph and without a dummy.*

Proof outline. The proof is a simultaneous strong induction on the vertex set. Call an ordered pair $x \to y$ bad when the stopped attacker wins from the defender state with queue (x, y) and untouched set $U = V \setminus \{x, y\}$. Put

$$\alpha = \deg_U(x), \qquad \beta = \deg_U(y),$$

let p_v be the full degree parity, and put $b_v = \sum_{w\sim v} p_w$. Three explicit zero-charge absorbers—a direct double close, one inserted open followed by three closes, and the complete zero-charge open fan—give, for every bad pair,

$$\beta = 1, \qquad b_x = 1, qquad\alpha = 1 \Longrightarrow |V| = 1 \pmod 2.$$

The same parity calculation gives the moment identity

$$b_x + b_y = (1 + \alpha + \beta)|V| + [xy](\alpha + \beta).$$

If the empty-root theorem failed, every first defender open x would have a winning attacker reply y, so every vertex would have an outgoing bad arc. Give a vertex rank $2, 1, 0$ according as $(p_v, b_v) = (0, 1), (1, 1)$, or $b_v = 0$. The three displayed constraints and the moment identity show that every bad arc strictly decreases this rank. A finite directed graph of positive outdegree has a directed cycle, contradicting strict descent. The three absorbers and the rank argument are uniform in the smaller induced graphs, which closes the simultaneous induction. □

Corollary 5.7 (close-first root exclusion)**.** *On an isolated-dummy initial board, a close-first attacker cannot force score one from either seat.*

Proof. If the defender moves first, Theorem 5.6 is stronger. If the attacker moves first and opens v, the defender can open a second vertex w for which $\deg_{V\setminus\{v,w\}}(v) = 0$. When the once-punctured degree of v is even, take the isolated dummy if available; when it is odd, take a neighbour whose removal toggles the degree. The attacker's forced close of v is therefore neutral and leaves a clear defender checkpoint covered by Theorem 5.5. The one-vertex terminal case is immediate. □

Consequently every hypothetical odd counterstrategy at the isolated-dummy root contains a clear attacker node at which it selects an open although a close is legal. Eliminating a leafmost such deviation requires its complete defender ancestry; conditioned close-first contraction alone does not do so.

6 The sharp frontier

The first-seat orientation has an especially economical root normal form. Write s_{xy} for the state after the two legal initial moves $\mathrm{OPEN}(x), \mathrm{OPEN}(y)$, with $x \neq y$.

Proposition 6.1 (first-seat two-open Bellman form)**.** *Suppose the carrier has at least two vertices. The first physical player can force score zero from the initial root if and only if there is a vertex x such that it can force score zero from s_{xy} for every $y \neq x$. Consequently it is sufficient to prove that every ordered pair $x \neq y$ gives a winning state s_{xy}.*

Proof. At the empty root the first player controls the move and must open some x. The resulting queue is the protected singleton (x) and the untouched set is nonempty. The opponent therefore cannot close or pass and its legal moves are exactly the opens $y \neq x$. Bellman recursion now gives the displayed existential–universal condition. The converse splices the assumed winning continuation below every such second open. □

This normal form does not assert that the required x exists. Its main value is causal: in a hypothetical first-seat counterstrategy, every first open is a defender branch of one strategy, and the attacker selects one second opener $f(x) \neq x$ below each branch. Thus the selected ordered pairs form a fixed-point-free functional digraph with genuine common root ancestry. A cycle argument would be sufficient only if its prefix cancellation can be lifted through the branch-dependent continuation cosets; this is a specialized form of the same factor-extension problem below.

More precisely, attach to a selected arc $x \to f(x)$ the universal prefix $S_V(x) + S_V(f(x))$, where $S_V(v)$ is the full live star. The sources and targets of a directed cycle are the same multiset, so these prefixes sum to zero. If the cycle length is odd, any chosen continuation representatives lift to an odd affine point at the common root; the isolated-dummy augmentation lock then says that their sum has nonzero real-edge projection. If the cycle length is even, the same sum is only a homogeneous response direction. Thus neither parity contracts the counterstrategy. One must add an in-tree branch or another earlier sibling to obtain both odd augmentation and zero projected moment.

The whole functional digraph does not repair this by incidence alone. There is a fixed-point-free map on seven labels with two even cycles and genuine feeding in-trees, together with a graph having an isolated dummy, for which the graph functional evaluates every selected two-OPEN arc prefix to one. Consequently every odd arc subfamily—including arbitrary combinations of cycle and in-tree arcs—has nonzero real-edge projection. Any successful functional-digraph argument must use correlations among the continuation cosets, not only the source–target incidence matrix.

Elementary congruence of the alternating adjacency matrix is also an exact reduction boundary rather than a free invariance. FIFO legality is graph-independent, so an even strategy on G and an odd strategy on an elementary shear G' can be interacted to a common public terminal word and a common live-star moment D. Their opposite scores force the shear row functional to satisfy

$$\rho_{G,i,j}(D) = 1.$$

Thus a proof of root invariance under graph congruence would have to exclude or neutralize precisely these strategy-correlated moments; public-state equivariance alone yields the unit defect instead.

Opening the isolated dummy first is another tempting specialization, but it only relocates the selector. From the pair state with queue (d, y), for every remaining real z the neutral square

$$C_d; \mathrm{OPEN}(z) = \mathrm{OPEN}(z); C_d$$

reconverges at the same score-zero state with queue (y, z). If every such dummy-consumed real-pair state is odd-winning, the odd player responds to either defender move by taking the other side of the square. Consequently an even win at (d, y) already forces an even-winning consumed pair (y, z) for some z. The isolated front therefore supplies exact schedule transport but not the missing continuation selector.

The affine ledger makes the same boundary sharp. On an odd total carrier the complete legal fan C_d together with all remaining opens has even cardinality. After forgetting dummy-incident coordinates, the sum of its open prefixes is not zero but the full real star $S_V(y)$. Every odd response point at the pair state evaluates to one plus the current potential, whereas $S_V(y)$ evaluates to the potential itself. Thus no local response point can have that projected value. An additional affine point from earlier root ancestry is necessary even in the dummy-first specialization.

Assume for contradiction that the affine condition of Theorem 3.1 fails, and fix the resulting exact odd strategy. Among its score-zero nodes choose one of minimum remaining rank, and among those one with minimum untouched-carrier size. This strategy-relative choice repairs the ancestry defect in a global minimum-hot argument: every node used below remains an actual descendant of the same root policy.

Theorem 6.2 (strategy-relative bad-predecessor trichotomy)**.** *The chosen node s is attacker-controlled and has score zero. Its selected move is a charge-one close of a real front f; translating* *the selected continuation by one gives a neutral winning subtree. Moreover s has an immediate defender parent p, an exact root-to-p prefix, and the complete legal fan at p. Exactly one of the following shapes occurs.*

1. *The incoming move is a charged close. For some y, $Q_p = (y, f, Q_0)$, the score at p is one, and C_y has charge one.*
2. *The incoming move is an open from a protected singleton: $Q_p = (f)$, the ko bit is set, and opening x produces $Q_s = (f, x)$.*
3. *The incoming move is an open from a clear nonempty queue. The old front is f, its charge before the open is zero, the untouched carrier has even cardinality, and f is adjacent to every untouched vertex. Opening x removes one such neighbour and makes the selected C_f charged.*

The putative fourth case—a zero-charge predecessor close followed by the selected charged close—is impossible.

Proof. At a minimum-rank score-zero node, an attacker-selected open or pass would leave a smaller score-zero descendant. Hence the selected move is a charged close, and isolation makes its front real. The immediate-parent decomposition retains the full defender fan and its root prefixes. A charged incoming close gives the first case. If the incoming close were neutral, an OPEN sibling at the same remaining rank would still have score zero but a strictly smaller untouched carrier, contradicting the secondary minimum.

For an incoming open, coherence splits on the parent's ko bit. A set ko bit forces the singleton queue in the second case. In the clear case, apply the same minimum-rank selected-close argument to every alternative OPEN sibling. The old front therefore has charge one after deleting each possible opened vertex. The elementary flip identity forces its charge on the old carrier to be zero and makes it adjacent to every untouched vertex; handshaking then gives even carrier size. These deductions, including the strategy prefixes and neutral translated continuation, are kernel-checked in `FifoStrategyBadAncestry`. □

Theorem 6.3 (clear-OPEN exclusion)**.** *The third alternative of Theorem 6.2 cannot occur in an initial isolated-dummy odd strategy. Hence every strategy-relative minimal bad node has either charged-CLOSE-spike or protected-singleton ancestry.*

Proof. Let the clear parent have queue (f, r) and untouched carrier U. Its complete fan contains the neutral sibling C_f, which has score zero and the same remaining rank as the chosen bad node. Rank minimality forces r to be nonempty, say $r = (y, r_0)$, and forces the attacker in this sibling to select a unit close of y. Thus

$$c(y, U) = 1.$$

For every $z \in U$, the same parent fan also contains the score-zero sibling O_z. Its queue begins (f, y) and it has the same minimal rank. The attacker selects C_f; the following defender node universally contains C_y. If C_y had unit charge, those two closes would return to score zero at smaller rank. Therefore

$$c(y, U \setminus \{z\}) = 0 \qquad (z \in U).$$

Comparing the two displayed equations shows that y is adjacent to every $z \in U$. But Theorem 6.2 gives $|U| = 0$ in $\mathbb{F}_2$, so universality gives $c(y, U) = |U| = 0$, contradicting the first display. The exact same-root prefixes, sibling subtrees, and rank comparisons are kernel-checked in `FifoStrategyBadAncestryClear`. □

This completes the predecessor classification inside the actual counterstrategy. The two surviving cases still require the causal factor extension below; the theorem does not by itself select compatible continuations across their earlier siblings.

Independently of that remaining exclusion, the selected unit close has a useful strategy-relative consequence. Every strict descendant of its selected child lies on score sheet one; translating that subtree by one gives a strategy in which every transition is score-neutral. In a protected singleton fan, applying the same minimum-rank argument to every forced open $z \in U$ gives neutral tails with queue (z) and untouched set $U \setminus \{z\}$. The defender may close z immediately, so

$$\deg_{U\setminus\{z\}}(z) = 0 \qquad (z \in U).$$

Thus the graph induced by U is Eulerian. This is a genuine reduction, not a contradiction: a cone from x over any even Eulerian graph has exactly these local parities.

6.1 Why descendants do not suffice

Even under this candidate normal form, the proof cannot be finished inside the selected subtree.

Proposition 6.4 (odd-spike descendant obstruction)**.** *There is a legal odd–odd spike for which the selected child and every off-spine descendant continuation are scalar-neutral, but the selected relative edge moment is not generated by the descendant continuation spaces.*

Proof. Take real vertices y, x, a, isolated dummy d, and edges ya, xa. At the defender state

$$U = \{a, d\}, \qquad Q = (y, x), \qquad k = 0,$$

select C_y, C_x. Both closes have charge one, and the residual board is edgeless. On the off-spine reply OPEN(a), let the attacker play C_y; only d remains untouched, so all future charges vanish. On OPEN(d), let it play OPEN(a); again all future charges vanish.

In the universal real edge space, every selected-spike terminal has relative moment $ya + xa$, whereas every off-spine terminal has moment zero. All three even continuation spaces are zero. Hence the selected coefficient cannot be cancelled by descendants. An earlier defender sibling is necessary. □

Even a complete defender fan inside an actual target-one policy need not contract locally.

Proposition 6.5 (complete-fan obstruction)**.** *Let the real edges be* $01, 02$ *and let* d *be isolated. The zero-charge prefix*

$$O_3, O_d, C_3, C_d, O_0$$

reaches the defender state

$$U = \{1, 2\}, \qquad Q = (0), \qquad k = 1,$$

whose complete legal fan is O_1, O_2*. One attacker policy forces terminal score one below both children.*

Proof. After O_1, play C_0, whose charge is the edge 02; after O_2, play C_0, whose charge is 01. Once 0 closes, no edge remains between a queued and an untouched real vertex, so every later close has charge zero. □

The contradiction at the initial root must therefore couple this fan to strictly earlier defender siblings.

There is also no uniform bound on the number of response histories in a valid local certificate, even after quotienting by cuts.

Proposition 6.6 (causal simplex caps)**.** *For every $r \geq 4$, a reachable defender node under a fixed close-first attacker has terminal labels which project onto*

$$\mathcal{C}_r = \{\mathbf{1}\} \cup \{\mathbf{1} + e_i : 1 \leq i \leq r\} \subseteq \mathbb{F}_2^r.$$

The set contains neither zero nor three distinct elements summing to zero. Its unique nonempty linear dependence has support r when r is odd and $r+1$ when r is even. In particular, odd zero-moment certificate size is unbounded.

Proof. Take real vertices $0, v$, an r-set U, and isolated dummy d. Use the prefix

$$O_0, O_d, C_0, O_v, C_d,$$

after which $Q = (v)$, U is untouched, and the defender moves. Let the attacker subsequently close whenever possible. Apply T_0 and retain only the coordinates vi, $i \in U$. If the defender first closes v, the label is $\mathbf{1}$. If it first opens i, the attacker immediately closes v, so the label is $\mathbf{1} + e_i$. Later play cannot change these coordinates.

Write $a = \mathbf{1}$ and $b_i = \mathbf{1} + e_i$. A relation $\alpha a + \sum_i \beta_i b_i = 0$ gives, in coordinate j,

$$\alpha + \sum_i \beta_i + \beta_j = 0.$$

Thus all β_j equal one bit. The only nontrivial relation uses every b_i and uses a exactly when r is even. Its support is therefore the stated odd number. For $r \geq 4$, pairwise sums have weight one or two, while members of $\mathcal{C}_r$ have weight $r-1$ or r, excluding a zero-sum triple. □

Propositions 6.4–6.6 rule out a childwise common-coset induction, a fixed bounded-support certificate, and a proof that chooses one distinguished reply in each fan. They do not refute Conjecture 1.1; every displayed node has additional ancestry available at the root.

6.2 Why ordinary topology and dummy deletion do not fill the gap

The affine-flow language can suggest a boundary-of-boundary or oik argument. On the actual attacker-pruned object this suggestion is false for a simple reason. Retain each history occurrence, even when two histories have the same public state, and let T_S be the resulting finite rooted strategy tree. Give every oriented tree edge e its universal move vector $\lambda(e)$.

Proposition 6.7 (history-tree obstruction)**.** *The boundary map*

$$\partial_1 : C_1(T_S; \mathbb{F}_2) \longrightarrow C_0(T_S; \mathbb{F}_2)$$

is injective. More precisely, for every odd terminal chain c there is a unique edge chain x satisfying

$$\partial_1 x = [\mathrm{root}] + c,$$

and

$$\sum_e x_e \lambda(e) = \sum_h c_h D(h).$$

Consequently the FIFO affine target is a labelled cokernel condition on (∂_1, λ), not a consequence of $\partial^2 = 0$ on the history tree.

Proof. For an edge e, let T_e be the component below it. In any solution the coefficient of e is forced to be $\sum_{h\in T_e} c_h$ by summing the vertex boundary equation over T_e. These coefficients give a solution because every internal nonroot vertex occurs once from its incoming edge and once for each selected outgoing subtree, while the odd augmentation leaves the root coefficient one. The same subtree formula proves uniqueness and hence injectivity. Finally each selected edge is counted once for every selected leaf below it; interchanging the two sums gives the path-moment identity. □

Attaching commuting schedule squares does not repair the argument. Let m be the edge-space-valued one-cochain which assigns an OPEN its current live star and assigns CLOSE and PASS zero. In a legal diamond with clear front f and untouched z, the schedules

$$O_z; C_f \qquad \text{and} \qquad C_f; O_z$$

reconverge publicly but their move moments differ by the basis edge fz. Thus the square has curvature $\delta m = fz$, not zero. Moreover a history-dependent attacker may choose different continuations at the two occurrences of the reconvergent state. Quotienting them would identify different continuation cosets; retaining them removes the square. Even the unlabelled carrier need not be Eulerian: after an attacker OPEN O_f on four vertices, the defender has three OPEN alternatives O_a, O_d, O_e; a history-dependent attacker may select C_f immediately after each. The resulting pre-close carrier consists of three isolated frontier points and no pair-faces. Its augmented boundary is the root, not zero.

There is an independent obstruction to induction by erasing the isolated dummy. Deleting a dummy interval can create a ko-illegal adjacent CLOSE/OPEN pair. The unique local repair exchanges the pair, changes the universal moment by one real edge, and reverses which player controls the repaired OPEN. If different defender branches open different real vertices, their repaired histories can reach the same dummy-free attacker node while requiring different selected moves. Hence the repaired leaves do not belong to one deterministic target policy. The correct object is a policy-indexed mapping cone whose repair-edge defects must be filled by earlier decorated siblings—precisely the ancestry term in Conjecture 6.11.

6.3 First-return blocks and outcome-valued debt

Another natural attempt is to stop whenever the FIFO queue first returns to empty and induct on the untouched carrier. The bookkeeping is exact, but it does not reduce the continuation to one score bit.

Proposition 6.8 (first-return factorization)**.** *Let b be a nonempty legal prefix from an empty queue to the first later empty queue, and suppose its endpoint is nonterminal. If B is the set of vertices opened in b, then b has $|B|$ OPENs, $|B|$ CLOSEs, no PASS, and length $2|B|$. In particular, the same physical player moves at the two empty-queue endpoints.*

There is a prefix vector P_b such that every continuation h on the residual carrier satisfies

$$D(bh) = P_b + \iota_b D(h).$$

On a real edge uv, the coefficient of P_b is zero if both endpoints are residual, one if exactly one lies in B, and, if both lie in B, their interval-disjointness bit inside b.

Proof. Queue length changes by $+1$ on OPEN, by -1 on CLOSE, and not at all on PASS, so the endpoint equation gives equally many OPENs and CLOSEs. A legal PASS requires the untouched set to be empty, which remains true thereafter; the nonterminal endpoint therefore excludes PASS. Every opened vertex is opened once, giving the length and mover statements.

For the vector identity, inspect one real edge. Two residual endpoints contribute only in h. If just one endpoint belongs to B, it opens while the residual endpoint is live and closes before that endpoint opens, so the edge occurs exactly once in the prefix live-star sum. If both endpoints lie in B, their whole interval relation has already been decided inside b. These three cases are exactly the displayed definition of P_b. $\square$

For one fixed attacker strategy, truncate at all first returns. If the continuation coset below an exit b is $A_b = a_b + W_b$, Proposition 6.8 gives the exact recursion

$$A_{\mathrm{root}} = \operatorname{Aff} \bigcup_b \big(P_b + \iota_b A_b\big).$$

Thus first-return grouping merely repackages the factor-extension problem: the different exits carry different prefix decorations and different continuation cosets.

Even the apparently weaker scalar induction, "return with score zero while the dummy remains, or terminate with score zero", is false in both seat orientations.

Proposition 6.9 (dummy-retention obstruction)**.** *For K_3 together with an isolated dummy, the player seeking odd score can defeat the stated stopped target when that player opens the block. For K_4 together with an isolated dummy, the same is true when the player seeking even score opens the block.*

Proof. On $K_3 + d$, the odd player opens d. Up to symmetry the reply is O_a; play C_d. The even player now sees

$$U = \{b, c\}, \qquad Q = (a), \qquad s = 0.$$

The move C_a is an immediate neutral first return with the dummy consumed. After O_b, answer C_a, which has charge one. The replies are C_b, which has charge one and gives another neutral first return with the dummy consumed, or O_c, after which the queue drains terminally at score one. The O_c branch at the displayed state is symmetric. These are all legal moves.

On $K_4 + d$, let the even player open the block. If it first opens d, open a real vertex. At the resulting even turn, C_d followed by a real OPEN, or a real OPEN followed by C_d, reaches

$$|U \cap K_4| = 2, \qquad Q = (a, b), \qquad s = 0.$$

Here an even CLOSE/CLOSE gives a neutral first return with d consumed; a real OPEN is answered by C_a, reaching score one with one real vertex untouched and queue length two. A CLOSE then followed by CLOSE returns empty at score one, while opening the last vertex drains terminally at score one.

If instead the first move is real, open a second real vertex. From the even turn with two real vertices and d untouched, exchange the replies C_a and O_d: either order reaches two untouched real vertices, queue (b, d), and score zero. There, C_b, C_d is a neutral consumed-dummy return; a real OPEN followed by C_b reaches score one with one real vertex untouched and queue (d, c), from which closing d or opening the last vertex drains at score one. The remaining even option is to open a third real vertex before touching d; answer C_a. At score one, with one real vertex and d untouched and queue length two, two CLOSEs give an empty score-one state with d untouched, while opening d or the last real vertex is answered by opening the other and drains terminally at score one. Symmetry exhausts all choices. $\square$

The correct residual summary after a consumed-dummy return is already outcome-valued. For a no-dummy residual graph H, let $e(H) = (e_0, e_1)$, where e_s says that physical seat s can force terminal score zero from the score-zero empty root. If the same mover returns at current score c, designated seat s is safe exactly when

$$(c = 0 \ \wedge \ e_s) \quad \text{or} \quad (c = 1 \ \wedge \ \neg e_{1-s}).$$

For $c = 0$ this is the definition. For $c = 1$, score translation changes the target to one, and finite determinacy says that seat s can force one exactly when the opposing seat cannot force zero. Hence a valid block induction must carry at least this two-seat outcome pair, or more canonically the affine continuation coset. Dummy presence and residual cardinality alone do not close the recursion.

Even that four-valued root sheet is not contextual under an active dummy interval. Deleting the dummy reverses the controller of precisely the real events strictly between O_d and C_d. Thus a strategy interaction must use the no-dummy strategy for seat s outside the interval and for the opposite seat inside it. The root outcome pair does not ensure that the second strategy is winning at a state selected by the first.

This failure already occurs on three labels. Take the real edge 0–2 and isolated dummy $d = 1$. The dummy-deleted empty root on $\{0, 2\}$ is winning for either designated seat. The prefixes O_d, O_0 and O_0, O_d have the same untouched real vertex, the same real queue after erasing d, the same turn, and the same score. Nevertheless for one fixed seat the first state is even-winning and the second is odd-winning. Hence the complete root outcome sheet forgets both dummy order and the branchwise ownership needed for a splice.

The direct two-policy interaction has an equally exact boundary. Odd strategies for complementary designated seats at the same state can be played against one another: at each nonterminal node one selects a move and the other contains its universal child. The recursion ends at a terminal state lying in both trees, and both require score one. Thus this diagonal is compatible, not contradictory.

The genuinely contradictory target is instead a pair of odd strategies for the same designated seat at states

$$s \qquad \text{and} \qquad \mathrm{scoreTranslate}_1(\mathrm{turnTranslate}(s)).$$

Turn–score conjugacy converts the first into an even strategy at the second root. Dummy timing cannot manufacture this pair. An isolated-dummy OPEN or front CLOSE preserves score exactly. At a dynamic cross exit, the companion histories retain a one-front queue lag, so they cannot be turn–score conjugate even when a charged real CLOSE supplies a unit score defect. The same square carries the real-edge curvature attached to that lag. Therefore two-copy self-play again requires a strictly earlier universal sibling to remove the queue/edge defect and align the branch-dependent continuation cosets; it is another derivation of the causal factor-extension obligation, not a proof of the conjecture.

There is a dual conservation law behind this failure. For an odd family of children at one defender node, suppose the lifted ancestry-and-move prefixes have zero value under the separating graph. Then the sum of the child continuations has value one, even after adding an arbitrary homogeneous root direction. In the one-dimensional quotient this is simply

$$A_{\mathrm{root}} = \{1\}, \qquad W_{\mathrm{root}} = \{0\}.$$

Thus an odd zero-prefix fan transports the affine unit into its continuation residue; complete-fan and curvature identities alone cannot destroy it.

Nor does immediate fan completeness force a single favorable third sibling. There is an exact policy on an edgeless score-one state with front 0 and untouched set $\{1, 2\}$ in which the front-CLOSE child selects O_1, while the immediate O_1 child selects O_2 and the immediate O_2 child selects O_1. All three are children of the same universal node, but neither OPEN sibling selects the lag-removing CLOSE. The example is not an initial-root counterexample; it proves that the missing repair must be an odd incidence across correlated continuation cosets, rather than one more immediate sibling.

6.4 The exact factor interface

For a node q in one fixed attacker strategy, retain the affine continuation coset $A_q = a_q + W_q$ from Section 3, and let Γ_q be its root-to-q prefix moment, projected away from dummy-incident coordinates.

Proposition 6.10 (ancestry factor interface)**.** *Let $q_1, \dots, q_{2k+1}$ be an odd list of nodes in the same strategy tree. Choose $a_i \in A_{q_i}$ and $d_i \in W_{q_i}$. If*

$$\sum_i (\Gamma_{q_i} + a_i) = \sum_i d_i,$$

then the projected root affine response space contains zero.

Proof. Since d_i is an even continuation direction, $a_i + d_i \in A_{q_i}$. Lifting along the actual ancestry gives $\Gamma_{q_i} + a_i + d_i \in A_{\mathrm{root}}$. An odd sum of points of one affine space remains in that affine space, and the displayed factor identity makes this sum zero. □

Without restrictions on the nodes, the converse is tautological: one may use the root itself. The content is to construct the list causally from the complete defender fans exposed by Theorems 6.2 and 6.3. The local identities already prove the required equations for prefix cuts, fixed-front remainders, and close-first descendants. What remains is to lift those equations through the attacker-pruned child cosets while preserving odd augmentation.

Conjecture 6.11 (causal factor extension)**.** *For every separating attacker strategy, the odd–odd spike or protected singleton frontier of Theorems 6.2 and 6.3 admits an odd ancestry factor satisfying the displayed identity of Proposition 6.10, using descendant continuation directions together with strictly earlier defender siblings.*

Conjecture 6.11 is the remaining proof obligation for Conjecture 1.1. In the relative-spine language, if h is the selected minimal bad node, $b_h = \Gamma_h + a_h$, and M_h is the sum of all frontier continuation spaces and all even sums of off-spine decorated points, the assertion is simply

$$b_h \in M_h.$$

The separating graph pairs to one with b_h and annihilates M_h in a hypothetical counterexample. Theorems 6.2 and 6.3 leave exactly the two candidate ancestry patterns to be contradicted; Proposition 6.4 shows why the strictly earlier siblings cannot be omitted.

7 Formal and computational evidence

The game semantics and subsidiary results are formalized in Lean 4 [1] using Mathlib [6]. The file `formal/Ogdoad/Fifo.lean` defines the transition system, proves rank descent, determinacy, score translation, the singleton and edgeless bases, the conditioned close-first theorem, and the stopped empty-root theorem. It also states `FifoLinkingTheorem` as a proposition; that declaration is not a proof of the proposition.

The file `formal/Ogdoad/FifoNormalization.lean` proves the queue-cut potential, live-star evaluation, the general two-open sum and its even/odd reply-fan corollaries, exclusion of close-first odd strategies at the isolated-dummy root, and the minimum-hot singleton-wall normal form. Related modules formalize fixed-policy affine spaces and individual causal transports. These are subsidiary lemmas, not a single kernel-checked proof of Conjecture 1.1.

The modules `FifoHub`, `FifoEmptyQueue`, `FifoOutcome`, and `FifoOutcomeBlock` check the score-erased relabelling, live-star transport, exact block counts, four-valued debt involution, and circularity of stopping at already favorable complete-outcome exits. `FifoBlockInduction` checks a conditional strategy-splicing interface but does not prove its scalar/parity block premises. The modules `FifoTreeTrace` and `FifoQPotential` certify small obstructions to the raw complete-leaf sum and to pointwise queue-cut normalization. These negative interfaces narrow the required ancestry construction; they do not settle the root conjecture. The modules `FifoStrategyInteraction` and `FifoSelfPlay` kernel-check the two-policy diagonal and the score–turn-conjugate contradiction target. They also certify that neutral dummy moves lack the required score defect and that a one-front-offset pair cannot be conjugate; these are self-play boundary theorems, not an end-to-end contraction. `FifoSeparatorFlow` proves that an odd zero-prefix fan retains separator value one in its continuation residue, and `FifoThreeSiblingBoundary` gives the exact same-tree countermodel to the naive immediate third-sibling selector. `FifoFunctionalDigraphBoundary` shows that adding genuine feeding in-trees to the selected first-seat cycles does not make incidence sufficient: an isolated-dummy separator evaluates every selected arc prefix to one. `FifoRootCongruence` proves the common-public-trace interaction across graphs and the resulting unit elementary-congruence row defect; it does not prove root-value invariance. Finally, `FifoPairState` proves that every score-zero mover failure with an OPEN available forces a lower minimum-hot singleton wall with two real charged endpoints. `FifoStrategyBadAncestry` performs the minimum inside the supplied strategy, retains the exact root ancestry and complete parent fan, and kernel-excludes the neutral-CLOSE predecessor. `FifoStrategyBadAncestryClear` excludes the clear-OPEN predecessor by combining its CLOSE sibling with every OPEN sibling. The only surviving minimal-bad ancestries are the charged-CLOSE spike and protected singleton.

The executable oracle `experiments/linking_game.py` implements the same state transition. Exact minimax verifies Conjecture 1.1 for every isomorphism class through eight real vertices after adjoining the isolated dummy, for either defender seat. At eight real vertices this is 12,346 graph classes, enumerated with the nauty toolchain whose algorithms are described in [5]. The script also checks the live-degree identities and the explicit obstruction schedules in this paper.

A separate exact root search found the following sharp selector witness on real vertices $0, \dots, 8$ with isolated dummy d:

$$\begin{aligned} E = \{&01, 02, 03, 04, 05, 06, 07, 08, 13, 15, 25, 28, 34, 35, 36, 37, 38, \\ &46, 57, 67, 68, 78\}. \end{aligned}$$

After the attacker opens 0, each real reply $1, \dots, 8$ is losing under unrestricted exact minimax; only the dummy reply wins. Here 0 has even degree, every other real vertex has odd degree, and the dummy is its unique same-parity mate. The winning dummy child also has a direct certificate: the cells

$$\{0, d\},\ \{1, 2\},\ \{3, 8\},\ \{4, 5\},\ \{6, 7\}$$

have even edge count between every two distinct cells, so Theorem 5.1 applies. This is an exact finite witness, not a proof of the remaining root selector or of Conjecture 1.1.

The census is a finite theorem about the tested orders only. It neither constructs the causal factor extension nor bounds its support. Proposition 6.6 in fact proves that a uniformly bounded-support version is false.

8 Conclusion

The isolated-dummy FIFO problem is exactly an affine response problem in a binary edge space. The scalar score, interval disjointness, live-star sum, and front-cut filtration are four forms of the same invariant. Local reply fans contract, parity-cell and matching boards are solved, and close-first counterstrategies are excluded. Strategy-relative minimization now proves the complete immediate-parent trichotomy and eliminates both the neutral-CLOSE and clear-OPEN cases. The remaining reduction asks whether the resulting charged-CLOSE-spike or protected-singleton ancestry factor can always be extended.

The remaining difficulty is not another parity identity. It is causal: prefix cancellations must be lifted through branch-dependent continuation cosets, and explicit examples require several defender siblings and unbounded affine support. Ordinary chain cancellation has no cycle on the pruned history tree, schedule diamonds have nonzero curvature, and dummy deletion changes policy ownership. Proving the causal factor-extension conjecture, or constructing an arbitrary-order obstruction, is the complete current frontier.

References

[1] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In André Platzer and Geoff Sutcliffe, editors, *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635, Cham, 2021. Springer. doi: 10.1007/978-3-030-79876-5_37.

[2] Lenwood S. Heath and Arnold L. Rosenberg. Laying out graphs using queues. *SIAM Journal on Computing*, 21(5):927–958, 1992. doi: 10.1137/0221055.

[3] Pavel Klavík, Yota Otachi, and Jiří Šejnoha. On the classes of interval graphs of limited nesting and count of lengths. In Seok-Hee Hong, editor, *27th International Symposium on Algorithms and Computation (ISAAC 2016)*, volume 64 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 45:1–45:13, Dagstuhl, Germany, 2016. Schloss Dagstuhl–Leibniz-Zentrum für Informatik. doi: 10.4230/LIPIcs.ISAAC.2016.45.

[4] C. L. Mallows and N. J. A. Sloane. Two-graphs, switching classes and Euler graphs are equal in number. *SIAM Journal on Applied Mathematics*, 28(4):876–880, 1975. doi: 10.1137/0128070.

[5] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

[6] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381, New York, NY, USA, 2020. Association for Computing Machinery. doi: 10.1145/3372885.3373824.